VERIS

Two-Agent Autonomous Protection Architecture

Connected Business Agent + Predictive Risk & Protection Agent

How the two agents operate independently, share evidence, resolve conflicts, and turn signals into actionable insurance protection workflows.

Core idea Veris does not wait for an SME to remember to update insurance. One agent understands what is changing inside the business; the other understands what is changing around it. A shared evidence and decision layer determines what matters and what action is justified.

SME OPERATIONS	VERIS CONNECT	SHARED EVIDENCE	VERIS SENTINEL	RISK / ACTION

Source basis: the team's EUREKA document identifies the main remaining gaps as automatic business-state detection and a complete action loop, while preserving the existing evidence-first, deterministic policy engine.
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1. The product architecture

1. Normal SME Software	2. Internal Agent	3. Shared Intelligence	4. External Agent	5. Action

The SME continues working in the software it already uses. Veris Connect sits inside that operating environment. Veris Sentinel is a separate intelligence workspace that consumes external signals and the SME context. Both agents feed the same retrieval, evidence, and deterministic policy engine; neither agent is allowed to make the final consequential decision on its own.

Layer	What it contains	Main question
SME OPERATIONS	ERP / accounting / POS / payroll / inventory / e-commerce / notes	Internal business signals
VERIS CONNECT	Event + change detection	What changed inside?
VERIS SENTINEL	Weather / indices / external risk sources	What is changing outside?

SHARED RAG	Policy + business + external evidence	What does the evidence say?
DECISION ORCHESTRATOR	Evidence precedence + rules + calculations	What should Veris conclude?
ACTION LAYER	Monitor / alert / review / adviser / carrier	What happens next?

2. Agent 1 - Veris Connect

Purpose: understand the business from inside the systems the SME already uses. The agent is embedded into the normal operating software, so the SME does not have to remember to report changes to Veris.

It watches authorised business signals such as accounting, ERP, POS, payroll, inventory, e-commerce, internal records, meeting notes, and digital documents. Its output is a structured business event or fact - not a final insurance judgment.

Internal source	Change detected	Insurance relevance	Policy retrieval	Orchestrator	Alert / action

Example Inventory moves from S$100,000 to S$150,000 in the connected accounting system. Veris Connect records the change, links it to the relevant stock-protection rules, and asks the shared engine to evaluate the insurance impact.

3. Agent 2 - Veris Sentinel

Purpose: understand risks and emerging conditions around the SME from relevant external sources. It is not required to sit inside the daily operating software; the SME can inspect it from the Veris Intelligence Center.

Typical inputs include weather and disaster feeds, published construction/equipment cost indices, economic and sector indicators, infrastructure disruption data, and other credible external signals. Its output is a risk signal or predictive indication, not proof that the SME has suffered a loss.

External source	Signal detected	SME context match	Policy relevance	Orchestrator	Monitor / alert / review

Example A published flood warning affects the area around an insured location. Veris Sentinel flags elevated exposure and retrieves the relevant policy context, but it does not claim that the SME has flooded unless SME-specific evidence exists.

4. Why the two agents are deliberately separate

VERIS CONNECT	VERIS SENTINEL
• First-party / internal business state	• Third-party / external environment
• Detects actual business changes	• Detects risk signals and emerging threats
• Examples: inventory, headcount, branch, equipment	• Examples: weather, indices, external disruption
• Typically higher-confidence current-state evidence	• Usually predictive or contextual rather than SME-specific proof

This separation prevents the system from treating a prediction as a fact. A connected accounting record can establish a current business value; an external cost index can indicate future pressure but should not silently overwrite the current value. The evidence model preserves that distinction.

5. Shared RAG and evidence architecture

Both agents feed a common retrieval system. This is important: there should not be two independent policy engines. The same policy wording, schedule, endorsements, business evidence, and external source records must be retrievable by both agents.

Trigger	Retrieve policy clauses	Retrieve business evidence	Retrieve external evidence	Verify provenance	Build evidence set

6. RAG workflow - what actually happens

Step	Stage	What happens
1	Trigger	An internal change or external risk signal is created.
2	Classify	Determine event/risk type: inventory change, new location, flood risk, cost pressure, etc.
3	Query plan	Generate targeted retrieval queries instead of searching every document indiscriminately.
4	Retrieve	Fetch relevant policy clauses, schedules, endorsements, business records, and external evidence.
5	Verify	Preserve document, page, source, timestamp, and text span. Reject

		fabricated or ungrounded quotes.
6	Grade	Mark evidence as PROVEN, DECLARED, or UNESTABLISHED according to source and authority.
7	Compute	Apply deterministic policy rules, arithmetic, dates, thresholds, and precedence.
8	Decide	The orchestrator assigns the final state: monitor, watch, review, action required, or potential incident.

7. The Decision Orchestrator - where the intelligence comes together

The two agents should not independently make competing final decisions. They produce structured signals. The shared Decision Orchestrator reconciles them using evidence provenance, source authority, timing, policy rules, and deterministic logic.

Connect output	Sentinel output	Policy evidence	Business evidence	Decision Orchestrator	Final state

8. Conflict resolution - simple rule

Rule 1 - Current business state The strongest SME-specific authoritative source determines the current business value. External predictions do not overwrite it.

Rule 2 - External risk External signals can increase attention or create a watch/review state, but do not prove SME-specific damage or loss.

Rule 3 - Agreement When internal evidence and external signals point in the same direction, priority increases because two independent signals reinforce the same concern.

Rule 4 - Uncertainty If evidence is insufficient, the system abstains and requests the missing evidence instead of guessing.

9. Conflict example

Suppose Hock Lee Foods has a policy with S$100,000 stock insurance.

Connect: stock = S$150k	Sentinel: cost index +10%	RAG: average clause	Orchestrator: separate current + future signals

Final output: Current documented stock = S$150,000. Current policy adequacy = 66.7%. External +10% cost index = forward-looking pressure, not a new stock value. The two signals strengthen the priority of a review, but the external signal cannot replace the internal source of truth.

10. What happens when the agents disagree?

Example: Connect says inventory is still S$100,000; Sentinel says industry prices suggest inventory values may have risen. Veris does not choose S$140,000 simply because a model predicted it. It keeps S$100,000 as the current documented value and creates a watch/review signal: "External cost pressure detected - current business value may need verification."

11. What the SME actually sees

The SME should not see two complex AI dashboards all the time. The normal business software remains the primary workspace, with subtle Veris interventions. A dedicated Veris Intelligence Center is available when the SME wants to understand what the agents are doing.

Normal SME software	Subtle Veris alert	Open insight	View evidence	Take action

12. Internal agent UI

Inside normal software, show contextual alerts such as: "Veris detected a 50% inventory increase." The SME can open the Veris Connect panel to see the source, timestamp, previous value, current value, insurance relevance, and next action.

13. External agent UI

In the dedicated Veris Intelligence Center, Sentinel shows the external environment: current risk level, relevant weather/events, cost-index movement, source timestamps, and which insured locations or business areas may be affected. This is a monitoring and intelligence experience, not another operational system.

14. Unified Protection View

The most important screen is where the two agents meet. Show three columns: BUSINESS STATE, EXTERNAL RISK, INSURANCE STATE. Under them, show the evidence-backed conclusion and the recommended next action.

BUSINESS STATE	EXTERNAL RISK	INSURANCE STATE	DECISION	NEXT ACTION

15. End-to-end autonomy

The agents should operate without requiring the SME to manually start an insurance review every time. Connect is event-driven from internal system changes; Sentinel runs scheduled or event-based checks against external sources. Both produce signals, the orchestrator decides whether the signal matters, and the action layer either continues monitoring, alerts the SME, or creates a human-review case.

16. Autonomy loop

Observe	Understand	Retrieve	Verify	Decide	Alert / Act	Record	Continue monitoring

17. Full example - one short scenario

Hock Lee Foods increases inventory from S$100,000 to S$150,000. Veris Connect detects the increase through the accounting system. At the same time, Veris Sentinel sees a 10% increase in the relevant published cost index. The shared RAG retrieves the policy average clause and the accounting evidence. The Decision Orchestrator keeps S$150,000 as the current documented value and treats the index change as forward-looking pressure. It calculates current adequacy at 66.7%, shows the relevant clause and creates a "Coverage Review Recommended" alert. The SME can share the case with the adviser, who receives the evidence and calculation in a Review Brief.

18. Adviser workflow

Case created	Review queue	Evidence + calculation	Request information	Adviser decision	Resolved / monitored

The adviser sees why the case exists, which agent generated each signal, which evidence supports it, what is missing, and what action is recommended. Veris supports the adviser; it does not replace licensed insurance judgment.

19. Carrier workflow

At carrier level, the same outputs can be aggregated for authorised Mode B portfolio intelligence. The key view is not a "violations list" but a Protection Attention Matrix: internal business drift on one axis, external risk on the other. High/high accounts receive the greatest attention.

Internal drift	External risk	Protection attention matrix	Priority accounts	Adviser action	Resolution rate

20. Final architecture in one diagram

SME Systems	VERIS CONNECT	Shared RAG + Evidence	DECISION ORCHESTRATOR	VERIS SENTINEL	External Sources

More precisely: the two agents are parallel sensing layers, not competing brains. The shared evidence/RAG layer grounds both. The deterministic Decision Orchestrator is the control point that resolves conflicts and turns signals into safe, explainable actions.

The pitch in one sentence Veris connects what is changing inside the business with what is changing around it, proves the insurance impact using the policy itself, and autonomously turns meaningful signals into timely protection actions.

21. Design principles to preserve

- No SME should have to remember to report every routine business change.
- Internal sources establish current business state; external sources provide context, risk signals, and forward-looking pressure.
- No agent may silently override stronger evidence.
- Every consequential finding must be traceable to source evidence.
- If evidence is insufficient, Veris must abstain.
- The same shared policy/RAG/rules engine powers both agents.
- The SME sees simple actions first and technical evidence only when needed.
- The adviser remains the authorised human decision-maker for regulated insurance actions.

Source note: This workflow is based on the team's EUREKA document and the agreed project direction. The source explicitly identifies automatic business-change detection, connected data feeds, action after detection, and the existing evidence-first deterministic engine as the key design context. ṆfileciteÖturn1file0ÖL247-L280Ộ